

LSIG: Long Semantic IDs for Generative Recommendation

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Abstract

Semantic ID-based generative recommendation models tokenize each item into a small number of discrete tokens and directly predict item identifiers from user interaction sequences. This approach has gained increasing traction in industrial applications. However, current methods are typically limited to short semantic IDs (e.g., just 3-4 tokens), which imposes a bottleneck on hierarchical expressiveness and hinders the accuracy and diversity of sequential recommendations. To overcome this limitation, we propose LSIG, a hierarchy-aware model tailored for long semantic ID modeling. LSIG consists of three main modules including 1) Token Personalized Attention (TPA), which dynamically aggregates prefix information from all levels of the long semantic ID, effectively mitigating semantic ambiguity in deep-layer representations; 2) Hierarchical Multi-Token Prediction (HMTP), which explicitly captures the hierarchical dependencies among semantic tokens and avoids the local optima caused by traditional token-by-token prediction; and 3) Prefix-Aware Loss (PA Loss), which suppresses training noise arising from semantically incorrect paths, thereby accelerating model convergence and enhancing generalization. Extensive experiments on the Alibaba App demonstrate that our framework consistently outperforms strong baselines in both offline evaluation and online business metrics.

CCS Concepts

• Information systems → Recommender systems.

Keywords

recommender system; Semantic ID; Generative Recall

ACM Reference Format:

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1 Introduction

Recommendation systems are fundamental to e-commerce platforms, social networks, and content delivery services. Their primary goal is to infer users' potential interests and deliver personalized item suggestions by modeling user-item interactions [3, 7, 23]. In recent years, sequential recommendation methods have demonstrated substantial performance gains by leveraging the sequential nature of user interactions [2, 5, 29, 39]. Representative models such as SASRec [15] and BERT4Rec [30] have shown the effectiveness of sequence modeling in capturing user intent dynamics. More recently, generative models have pushed the boundaries of recommendation by treating item prediction as a sequence generation task [1, 36, 38]. However, conventional ID-based representations often struggle to capture the rich semantic attributes of items, especially in cold-start and long-tail scenarios. To address this limitation, recent studies have explored semantic ID-based recommendation, where each item is represented as a sequence of discrete tokens that encode semantic structure [10, 25, 27, 38, 44, 45]. These semantic IDs are typically obtained via discrete representation techniques such as product quantization (PQ) or vector quantization (VQ), enabling token-by-token generation of items in a generative framework. For instance, TIGER [25] employs a residual quantization variational autoencoder (RQ-VAE) to encode item features into hierarchical semantic IDs, allowing semantically related items to share information. COBRA [37] further enhances interpretability and generalization by generating coarse-to-fine hierarchical semantic IDs from dense embeddings.

Despite these advances, current semantic ID-based generative models often use short token sequences to represent items. For example, TIGER adopts only four tokens per item. This token budget may be insufficient for encoding rich item semantics, especially for complex or information-rich products [16]. Short semantic IDs limit the model's capacity to distinguish fine-grained item attributes, resulting in information loss and degraded performance in complex recommendation scenarios. While increasing the semantic ID length seems a natural remedy, we observe a diminishing return in efficiency. As shown in Figure 1, extending the token sequence

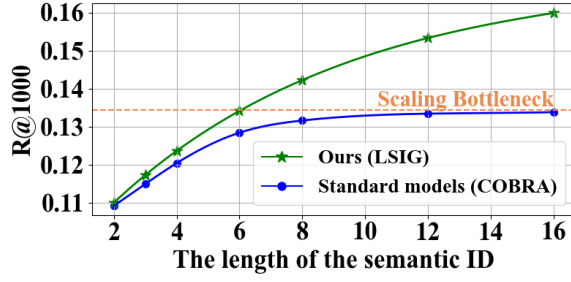


Figure 1: Merely increasing semantic ID length leads to a bottleneck in model efficiency. Our proposed LSIG significantly improves the scalability of semantic encoding.

length increases training cost (in FLOPs) without proportional performance gains, exposing a scaling bottleneck.

In this work, we analyze the underlying causes of the semantic ID scaling bottleneck and propose LSIG, a hierarchy-aware generative recommendation framework designed to more effectively leverage long semantic ID sequences. Developing such a model presents several key challenges:

(1) **Semantic ambiguity caused by prefix-sensitive tokens.**

In long hierarchical semantic IDs, each token encodes part of the item’s attributes in a coarse-to-fine manner. Since each token’s semantics depend heavily on its preceding prefix, items with different prefixes but identical suffix tokens may exhibit inconsistent semantics. Feeding such discrete token sequences directly into the model can lead to ambiguity and degraded generation quality. Therefore, static token representations are insufficient to capture the contextual dependencies required for disambiguation. (2) **Lack of explicit modeling of hierarchical dependencies.** Most existing methods adopt a next-token prediction paradigm, which generates tokens greedily in a left-to-right manner. This stepwise strategy fails to capture the global dependencies across token layers. As a result, prediction errors in early tokens tend to propagate downward, leading to structurally invalid or incoherent semantic IDs. (3) **Uneven semantic importance across token positions.** In long semantic ID sequences, early tokens usually encode high-level category semantics that dominate the overall meaning, while later tokens carry fine-grained details. Uniformly optimizing all token positions may cause the model to overfit to less informative tail tokens, thus undermining its ability to learn the core semantic structure.

To address these challenges, we introduce LSIG, a framework with three key innovations:

- **Token Personalized Attention (TPA).** To dynamically refine each token’s representation. By leveraging item category embeddings and prefix tokens, TPA mitigates semantic ambiguity caused by prefix variations.
- **Hierarchical Multi-Token Prediction (HMTP).** We propose an HMTP objective that synchronously predicts multiple subsequent tokens, combined with a Throttling Gate Unit to explicitly model the hierarchical dependencies within semantic IDs.

- **Prefix-Aware Loss (PA Loss).** To emphasize the semantic backbone of the ID, we introduce a prefix-aware loss function that dynamically truncates the loss computation to the longest correctly predicted prefix.

2 Related Work

2.1 Sequential Recommendation

In recent years, the generative paradigm has attracted significant research interest in recommendation systems. A representative approach models the recommendation task as a sequence generation problem, employing Transformer-based autoregressive architectures to generate personalized target item sequences from users’ historical interactions, thereby enabling end-to-end modeling of recommendations [4, 8, 14, 32, 38]. SASRec [15] performs autoregressive prediction using unidirectional self-attention mechanisms, whereas BERT4Rec [30] employs bidirectional Transformers to enhance contextual modeling capabilities. The industrial deployment of GenRank [14], MTGR [8], and HSTU [38] has also demonstrated the practical advancement of sequential recommendation techniques.

However, directly using spares, dynamically changing item IDs as model input tokens presents two major challenges: (1) IDs inherently exhibit sparsity and lack semantic clarity, hindering effective sequence pattern learning; (2) models demonstrate limited generalization capabilities for Cold-start items. To address these issues, researchers have explored encoding item modal features as discrete tokens (semantic IDs) [25]. This approach has driven the development of semantic ID-based sequential recommendation methods.

2.2 Semantic ID-based Recommendation

Semantic ID represents a sequence structure composed of discrete tokens that capture the semantic attributes of items. These tokens often possess hierarchical structures and are encoded through feature extraction from multiple modalities (e.g., text, images, interactions) [10, 17, 25, 28, 35, 41]. Construction methods include vector quantization [25, 34, 42, 43, 45, 47], hierarchical clustering [13, 17, 26], and joint training with recommendation models [19]. Compared to sparse item IDs, semantic IDs offer compact vocabularies, enhanced transferability and interpretability, and improved model scalability and inference efficiency [20, 24, 27, 38, 45].

With the advancement of generative modeling, semantic IDs have increasingly been employed as token-level generation targets. Models generate each token sequentially to compose the semantic ID of candidate items. TIGER [25] pioneered the complete implementation of semantic ID generation, utilizing RQ-VAE for hierarchical quantization of item content to enable token-level generative retrieval. COBRA [37] introduced a two-stage generation framework, first generating semantic sparse IDs and subsequently refining them into semantic dense vectors to facilitate coarse-to-fine retrieval. Recent work such as OneRec [4] has further utilized semantic IDs to unify retrieval, ranking, and generation tasks within a single architecture, enhancing model versatility and training efficiency. However, these approaches primarily focus on performance improvement through additional modalities or mechanisms, without thoroughly investigating the potential of models to exploit semantic encoding structures.

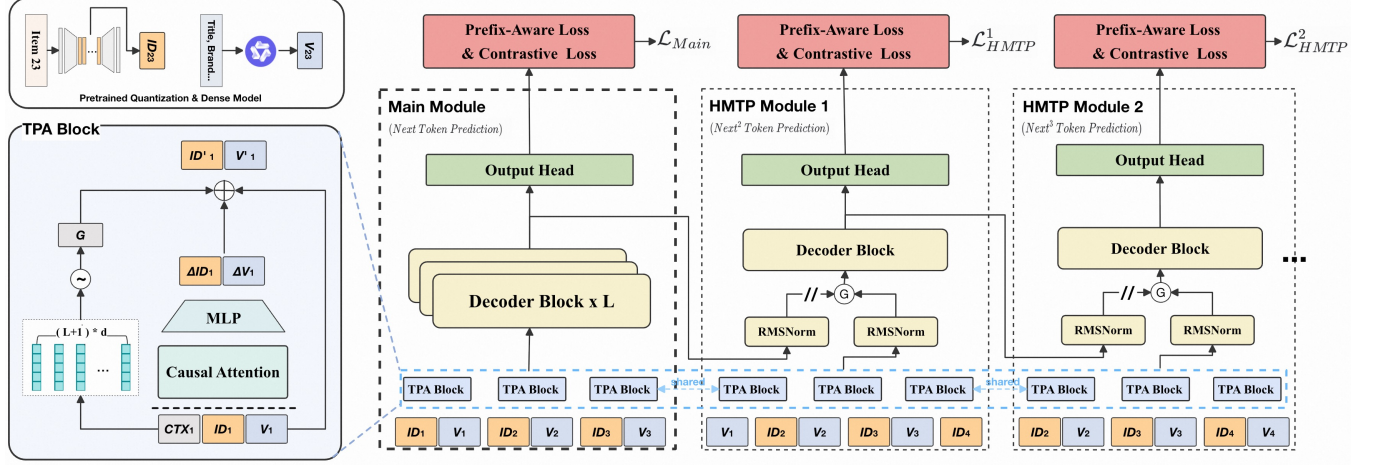


Figure 2: The overall framework of LSIG.

3 Preliminaries

In this section, we introduce the problem definition, the basic concepts, and the notations in this paper.

3.1 Problem Definition

We formulate this task as a sequential recommendation problem: Given a user's historical interaction sequence:

$$s = \{i_1, i_2, \dots, i_{t-1}\} \quad (1)$$

where i_k denotes the user's interaction with an item at time step k , the objective is to predict the next potentially relevant item i_t :

$$\hat{i}_t = \arg \max_{i \in \mathcal{I}} P(i | s) \quad (2)$$

where $\mathcal{I}$ denotes the set of all candidate items, and $P(i | s)$ represents the conditional probability distribution over items given the interaction history s . Unlike conventional approaches that rely on dense item IDs, this work introduces quantified semantic IDs as generation targets to explicitly model hierarchical semantic structures.

3.2 Quantified Semantic ID

Traditional item IDs are semantically meaningless discrete symbols, making it difficult to capture the structural relationships among items. To address this limitation, we adopt a discrete semantic representation mechanism that encodes each item as a semantic ID composed of multiple latent tokens. The construction of this representation involves the following steps:

- We apply a residual quantization variational autoencoder (RQ-VAE) to hierarchically quantize the high-dimensional feature vector of each item;
- At each quantization level, an index token is selected from a corresponding codebook, forming a hierarchical discrete token sequence, denoted as:

$$ID_t = \{z_t^{(1)}, z_t^{(2)}, \dots, z_t^{(L)}\} \quad (3)$$

where $z_t^{(l)} \in \mathcal{Z}^{(l)}$ denotes the token selected at the l -th layer, and $\mathcal{Z}^{(l)}$ is the codebook associated with that layer. This hierarchical

semantic ID not only preserves the semantic characteristics of items but also encodes their multi-level structural relationships, laying a solid foundation for downstream recommendation tasks.

3.3 Transformer Decoder

We employ a Transformer-based autoregressive decoder architecture. Each decoder block contains a multihead self-attention module, two add-and-norm modules, and a feed-forward network [33]. The decoding process is defined as:

$$\hat{ID}_t = \text{Decoder}(ID_{<t}) \quad (4)$$

where $\hat{ID}_t^{(l)}$ denotes the predicted semantic ID for the t -th item. The model autoregressively generates the next item's semantic IDs in a token-by-token manner.

4 Methodology

This section presents our proposed LSIG framework, which enables efficient and coherent generation of semantic IDs by explicitly modeling and leveraging the hierarchical semantics embedded in long semantic identifiers. (1) We begin by introducing how **Token Personalized Attention (TPA)** transforms each item's sparse semantic ID into a continuous representation to mitigate token-level semantic ambiguity. (2) Next, we propose a **Hierarchical Multi-Token Prediction (HMTF)** mechanism that improves the model's ability to capture hierarchical dependencies within token sequences, thereby enhancing generation coherence. (3) We then design a **Prefix-Aware Loss (PA Loss)** that guides the model to optimize semantic ID generation in a coarse-to-fine manner, significantly boosting training efficiency and generalization capability. (4) Finally, we analyze the **scalability of long semantic ID representations** under different encoding depths (i.e., scaling laws), offering both theoretical insights and practical guidance for designing more expressive semantic encoding structures. The overall architecture of LSIG is illustrated in Figure 2.

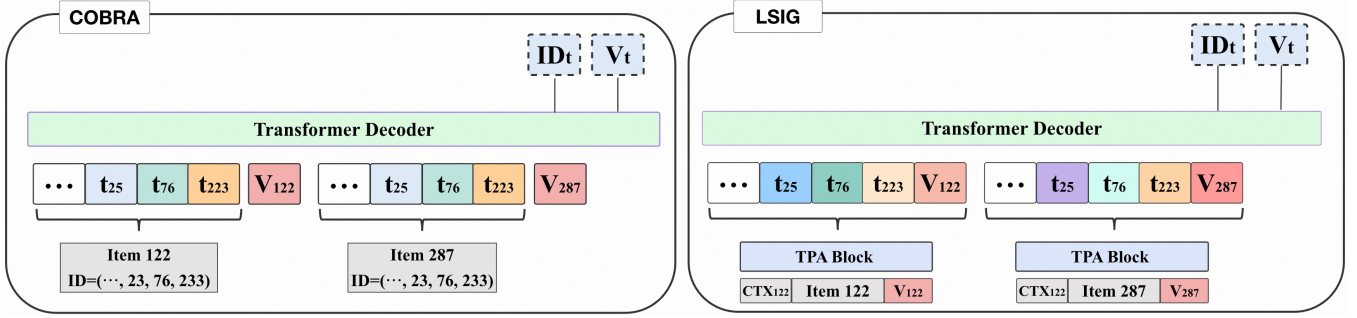


Figure 3: Comparison of Input Structures between COBRA(left) and TPA(right). The left part illustrates the conventional semantic ID input used by COBRA, where sparse semantic IDs and dense vectors V are directly fed into the decoder. In contrast, the right part depicts our proposed TPA architecture, which introduces a prefix-aware continuous refinement module before the main model. This module dynamically adjusts token representations by capturing contextual semantic dependencies between each token and its prefix, while preserving the clustering structure of semantic IDs.

4.1 Long Continuous Semantic ID Representation

4.1.1 Long Sparse Semantic ID Representation. LSIG adopts Residual Quantization Variational Autoencoder (RQ-VAE), inspired by TIGER [25], to construct sparse semantic ID sequences. To preserve fine-grained semantic variations and evaluate the representational capacity of long semantic codes, we extend the semantic ID length to 16 layers, significantly exceeding the typical depth of 3 to 4 layers used in prior methods[10, 25, 28]. Moreover, following COBRA [37], we enrich item representations by incorporating both dense image features and textual content. The contrastive learning loss is also aligned with COBRA:

$$\mathcal{L}_{\text{dense}} = \log \frac{\exp(\cos(\hat{\mathbf{v}}_{t+1} \cdot \mathbf{v}_{t+1}))}{\sum_{\text{item } j \in \text{Batch}} \exp(\cos(\hat{\mathbf{v}}_{t+1}, \mathbf{v}_{\text{item } j}))} \quad (5)$$

4.1.2 Semantic Ambiguity of Sparse IDs. Because each token in the codebook is highly dependent on its prefix path, directly feeding sparse semantic IDs into the model may lead to semantic ambiguity. For instance, consider two token sequences: $\{\text{Semantic ID} \in (\dots, 3, 16, 24, 55)\}$ and $\{\text{Semantic ID} \in (\dots, 7, 16, 24, 55)\}$. Although they share the same suffix(16, 24, 55), their prefix paths ($3 \rightarrow 16$ vs. $7 \rightarrow 16$) encode entirely different semantics, as they belong to distinct high-level categories (Figure 4). These similarities often arise from fine-grained attribute overlap (e.g., similar patterns or appearances), causing the same tail tokens to appear in different semantic contexts.

If the model treats such identical tokens equally, regardless of their prefix, it may incorrectly conflate distinct semantic meanings, thereby harming the quality of recommendations. To resolve this issue, we propose the Token Personalized Attention (TPA) module.

4.1.3 Token Personalized Attention (TPA). As shown in Figure 3, before being fed into the main model, token sequences are fine-tuned by the Token Personalized Attention (TPA) module in a prefix-sensitive manner to alleviate semantic ambiguity among identical tokens.

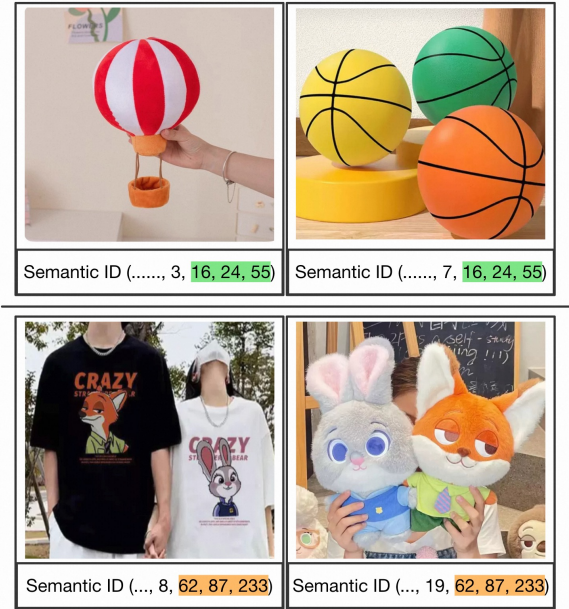


Figure 4: Fine-grained similar items tend to share similar suffix encodings.

Specifically, for each TPA Block, we input a set of semantic ID representations for items, including the discrete token representation $ID_t \in \mathbb{R}^{L \times d}$ and the corresponding dense attribute embedding $V_t \in \mathbb{R}^d$, where L denotes the length of the semantic ID, and $t \in \{1, 2, \dots, T\}$ represents the t -th item in the user behavior sequence. We first apply causal attention to extract prefix-relevant features and obtain fine-tuned token representations. Furthermore, to enhance the ability of suffix tokens to capture category information embedded in prefix tokens, we incorporate the category context $CTX_t \in \mathbb{R}^d$ into the first position of the TPA Block input.

$$[\Delta ID_t; \Delta V_t] = \text{MLP}(\text{Attn}_{\text{causal}}[CTX_t; ID_t; V_t]) \quad (6)$$

where MLP and $Attn_{causal}$ denote multi-layer perceptron and causal attention, respectively. Next, we compute the fusion gate G_t for each token using a learnable projection vector $w_m \in \mathbb{R}^{(L+1)*d}$:

$$G_t = \text{Sigmoid}(w_m^T [CTX_t, ID_t; V_t]) \quad (7)$$

Finally, the fine-tuned token representation is computed via a gated fusion between the original input and the updated features:

$$[ID'_t; V'_t] = G_t \cdot ([ID_t; V_t]) + (1 - G_t) \cdot ([\Delta ID_t; \Delta V_t]) \quad (8)$$

The TPA module, by introducing a category-aware prefix adjustment mechanism, effectively mitigates semantic ambiguity of tokens and enhances the fine-grained expressiveness of semantic IDs.

4.2 Hierarchical Multi-Token Prediction

4.2.1 Semantic ID Consistency. Semantic IDs, constructed through discrete encoders such as RQ-VAE, inherently form hierarchical semantic sequences that encode item attributes from coarse to fine levels. However, existing generative recommendation models commonly adopt a next-token prediction paradigm, which overlooks the structural dependencies among tokens. This design leads to two critical limitations:

- **Error Accumulation:** Since tokens are generated sequentially, any misprediction in an earlier layer can propagate and amplify through subsequent layers, ultimately yielding invalid or semantically inconsistent token sequences (e.g., meaningless or structurally broken codes);
- **Structural Inconsistency:** Greedy token-wise generation focuses on local optimization without modeling the global semantic structure of the ID. This can disrupt the coherence of the underlying codebook and degrade the expressiveness of the generated representations.

4.2.2 Hierarchical MTP Modules. To address these issues, inspired by Multi-Token Prediction (MTP) [6, 18], we introduce a Hierarchical Multi-Token Prediction (HMTP) mechanism that explicitly captures the hierarchical dependencies across semantic layers. Unlike RPG [12], which predicts all tokens in parallel, HMTP module maintains a complete causal chain by predicting tokens sequentially, where each level's prediction is constructed based on the contextual features of the previous level.

Specifically, at prediction depth k , for the i -th input token t_i , we utilize the throttling gate to combine the previous hidden state $h_i^{k-1} \in \mathbb{R}^d$ and the TPA embedding of the target token $TPA(t_{i+k}) \in \mathbb{R}^d$, as follows:

$$h_i'^k = G_k(\text{RMSNorm}(h_i^{k-1}); \text{RMSNorm}(TPA(t_{i+k}))) \quad (9)$$

As shown in Figure 5, the gating function $G_k(\cdot)$ is designed to restrict cross-item information flow, ensuring causal consistency during online inference. Concretely, when the predicted depth t_{i+k} exceeds the maximum semantic ID length L of the current item (i.e., when t_i belongs to the n -th item, and t_{i+k} crosses into item $n+1$), the hidden state h_i^{k-1} is suppressed, and only the teacher-forcing

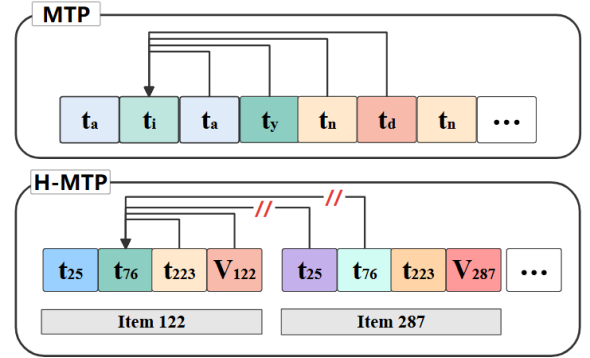


Figure 5: Illustration of the training differences between the MTP layer and the HMTP layer. The arrows indicate the direction of gradient backpropagation during training. By constraining gradient propagation across items, HMTP ensures consistency between training and inference.

token embedding t_{i+k} is preserved:

$$G_k(\cdot) = \begin{cases} \alpha \cdot h_i^{k-1} + (1 - \alpha) \cdot t_{i+k}, & \text{if } \exists n \in \mathbb{N}, nL < i \wedge (n+1)L > i+k \\ t_{i+k}, & \text{otherwise} \end{cases} \quad (10)$$

where $\alpha \in [0, 1]$ is a fusion hyperparameter. This gating mechanism dynamically restricts token-level fusion to the same item segment, thus aligning training-time supervision with inference-time causality. Note that the TPA layer is shared across all HMTP modules and the main decoder. The resulting fused representation $h_i'^k$ is subsequently passed through the k -th Transformer block to produce the depth-specific output h_i^k :

$$h_i^k = \text{TRM}_k(h_i'^k) \quad (11)$$

In summary, Hierarchical MTP explicitly encodes the inter-token dependencies across semantic ID levels, allowing the model to leverage richer structural context throughout generation. Together with the boundary-aware Throttling Gate mechanism, LSIG ensures consistent behavior across training and inference stages, thereby improving stability, generalization, and reasoning coherence.

4.3 Prefix-Aware Training Objective

4.3.1 Observation. In our hierarchical tokenization scheme based on RQ-VAE, each item is represented by a semantic ID composed of discrete tokens drawn from multiple codebooks. These tokens are arranged in a hierarchical order, from coarse-grained to fine-grained semantic levels, naturally encoding a multi-level dependency structure. Within this hierarchy, the correctness of higher-level (i.e., coarse-grained) tokens plays a crucial role in determining the semantic validity of the entire ID. For example, if the first-level category token is predicted incorrectly, even a perfect prediction of all subsequent fine-grained tokens may still yield an inconsistent or semantically misleading ID. Therefore, we posit that coarse-level tokens carry greater semantic importance than fine-level ones, and that the training objective should reflect this inherent asymmetry.

Table 1: Performance comparison between baseline methods and LSIG. The best results are highlighted in bold, while the second-best results are underlined. LSIG statistically outperforms the best baselines at a significance level of $p < 0.05$ by paired t-test.

Model	Sports and Outdoors				Beauty				Ali_Rec			
	R@5	N@5	R@10	N@10	R@5	N@5	R@10	N@10	R@100	N@100	R@1000	N@1000
<i>Item ID-based</i>												
Caser	0.0116	0.0072	0.0194	0.0097	0.0205	0.0131	0.0347	0.0176	0.0373	0.0097	0.0944	0.0160
HGN	0.0189	0.0120	0.0313	0.0159	0.0325	0.0206	0.0512	0.0266	0.0419	0.0115	0.1027	0.0194
GRU4Rec	0.0129	0.0086	0.0204	0.0110	0.0164	0.0099	0.0283	0.0137	0.0395	0.0109	0.0970	0.0163
BERT4Rec	0.0115	0.0075	0.0191	0.0099	0.0203	0.0124	0.0347	0.0170	0.0398	0.0111	0.0991	0.0181
FDSA	0.0182	0.0122	0.0288	0.0156	0.0267	0.0163	0.0407	0.0208	0.0412	0.0114	0.1001	0.0185
SASRec	0.0233	0.0154	0.0350	0.0192	0.0387	0.0249	0.0605	0.0318	0.0437	0.0119	0.1073	0.0196
S ³ -Rec	0.0251	0.0161	0.0385	0.0204	0.0387	0.0244	0.0647	0.0327	0.0445	0.0120	0.1089	0.0199
<i>Semantic ID-based</i>												
VQ-Rec	0.0213	0.0147	0.0305	0.0177	0.0467	0.0321	0.0679	0.0393	0.0477	0.0129	0.1149	0.0212
TIGER	0.0267	0.0186	0.0406	0.0229	0.0462	0.0327	0.0661	0.0391	0.0498	0.0135	0.1201	0.0221
COBRA	<u>0.0310</u>	<u>0.0218</u>	<u>0.0438</u>	<u>0.0261</u>	<u>0.0543</u>	<u>0.0404</u>	<u>0.0739</u>	<u>0.0466</u>	<u>0.0535</u>	<u>0.0153</u>	<u>0.1349</u>	<u>0.0255</u>
LSIG	0.0346	0.0243	0.0495	0.0295	0.0581	0.0435	0.0810	0.0508	0.0630	0.0176	0.1599	0.0295

4.3.2 Prefix-Aware Loss Function. To emphasize the hierarchical importance of token positions, we introduce a Prefix-Aware Loss that dynamically identifies the longest correctly predicted prefix of a semantic ID and only computes loss up to and including the first incorrect token. This approach guides the model to prioritize learning coarse-level token prediction before attending to finer-grained layers. Formally, for each item, let the ground-truth semantic ID be: $t = [t_1, t_2, \dots, t_L]$ and the predicted semantic ID be: $\hat{t} = [\hat{t}_1, \hat{t}_2, \dots, \hat{t}_L]$. We define the longest correctly predicted prefix as:

$$\ell = \max \{j \in [0, L] \mid \forall i \leq j, \hat{t}_i = t_i\} \quad (12)$$

The Prefix-Aware Loss is then computed as the cumulative cross-entropy over the first $\ell + 1$ positions:

$$\mathcal{L}_{PA} = \sum_{i=1}^{\ell+1} \text{CrossEntropy}(\hat{t}_{i+1}, t_{i+1}) \quad (13)$$

By dynamically adjusting the optimization scope during training, this loss avoids unnecessary updates to deep-layer tokens when upper-layer predictions are incorrect. It effectively suppresses gradient noise from semantically invalid paths, accelerates convergence, and ensures that the model first captures the high-level semantic structure before learning finer details.

4.3.3 Prefix-Aware Training in HMTP and Inference Decoupling. In LSIG, we apply the Prefix-Aware Loss to supervise each layer of the Hierarchical Multi-Token Prediction (HMTP) module. Specifically, for the k -th layer of HMTP and the t -th item in the user behavior sequence, we compute the corresponding prefix-aware loss $\mathcal{L}_{PA}^{(k,t)}$. The overall HMTP loss is defined as:

$$\mathcal{L}_{MTP} = \frac{\lambda}{H} \sum_{k=1}^H \sum_{t=1}^T \mathcal{L}_{PA}^{(k,t)} \quad (14)$$

where H denotes the number of HMTP layers, T is the number of items in the user sequence, and λ is a scaling hyperparameter for balancing the loss. This objective enforces semantic coherence across token levels during training. During inference, we decouple HMTP from the main decoder and rely solely on the base model for token generation, thus ensuring efficient deployment without sacrificing training benefits.

5 Experiments

In this section, we empirically demonstrate the effectiveness and efficiency of the proposed LSIG method.

5.1 Experiment Setup

5.1.1 Datasets and Evaluation Metrics. We conduct experiments using two categories from the Amazon Reviews dataset [22]: “Sports and Outdoors” (Sports) and “Beauty” (Beauty), which are commonly used for benchmarking semantic ID-based generative recommendation methods [12, 25, 37]. To further evaluate scalability in real-world settings, we additionally use Ali_Rec, a large-scale industrial dataset from the Alibaba e-commerce platform containing over 168 million recommendation sessions. We adopt a dual-perspective evaluation to assess both ranking accuracy and real-world recommendation impact: 1) **Offline Evaluation:** Following the setup in Rajput et al. [25], we evaluate performance on Recall@K and NDCG@K with $K \in \{5, 10, 100, 1000\}$, using the best-performing checkpoint on the validation set. 2) **Online Metrics:** The primary evaluation metrics include conversion rate and Gross Merchandise Volume (GMV), which directly reflect user engagement and economic impact.

5.1.2 Baselines. We compare LSIG with a variety of representative baselines, including both item ID-based and semantic ID-based methods:

- Caser [31]: Employs convolutional filters to capture local patterns in item ID sequences.
- HGN [21]: Integrates gating mechanisms within an RNN-based model.
- GRU4Rec [9]: A GRU-based model tailored for session-based recommendation.
- BERT4Rec [30]: Utilizes a bidirectional Transformer encoder trained with a Cloze-style objective.
- FDSA [40]: Processes item ID and feature sequences independently via self-attention blocks.
- SASRec [15]: Leverages a Transformer decoder with binary cross-entropy loss to model item ID sequences.
- S³-Rec [46]: First pre-trains with self-supervised tasks to align item IDs and features, then fine-tunes on the next-item prediction task.
- VQ_Rec [11]: VQ_Rec employs product quantization to tokenize items into semantic IDs, which are then aggregated through embedding pooling to form the final item representations.
- TIGER [25]: Employs RQ-VAE for item quantization and generates semantic IDs autoregressively using beam search.
- COBRA [37]: A cascaded generation process is employed to alternately produce sparse semantic identifiers and dense representations, with inference facilitated by BeamFusion search.

5.1.3 Implementation Details. LSIG is trained using the Adam optimizer with an initial learning rate of 0.0016 and a batch size of 256. All experiments are conducted on NVIDIA H20 GPUs with 80GB of memory. We perform grid search for hyperparameter tuning and report the most optimal hyperparameter settings. Specifically, the model consists of $D = 2$ decoder blocks, and $H = 4$ HMTP layers. The semantic ID space is constructed using a codebook depth of $L = 16$, with each layer containing $W = 256$ codewords. We use the most recent $T = 50$ historical behaviors as user context during training and inference. For generation, beam search is employed with a width of $N_S = 10$ sequences per inference.

5.2 Experimental Results and Discussion

5.2.1 Offline Evaluation. We benchmark LSIG on the Ali_Rec dataset against both item ID-based and semantic ID-based baselines. Results are summarized in Table 1. Semantic ID-based methods consistently outperform traditional item ID-based baselines, highlighting the benefit of leveraging hierarchical semantics. Among all baselines, LSIG achieves the overall best performance, surpassing the strongest competitor by +18.53% in Recall@1000 and +15.69% in NDCG@1000 on average. As illustrated in Figure 1, LSIG also substantially improves the scalability of semantic encoding. To ensure a fair comparison, all semantic ID-based methods are evaluated with a fixed semantic ID length of 16 tokens.

5.2.2 Ablation Study. To disentangle component contributions, we conduct ablation studies, summarized in Table 2, and draw three key observations: 1) Effectiveness of TPA: Removing TPA results in significant performance drops, validating the hypothesis that sparse semantic encoding introduces ambiguity in token semantics. 2) HMTP for Inference Consistency: Disabling HMTP impairs

Table 2: Ablation analysis of LSIG.

Variants	R@100	$\Delta(\%)$	N@100	$\Delta(\%)$
Full Model	0.0630	-	0.0176	-
Ours w/o TPA	0.0574	-8.89%	0.0164	-6.82%
Ours w/o HMTP	0.0596	-5.40%	0.0165	-6.25%
Ours w/o PA Loss	0.0603	-4.29%	0.0171	-2.84%

structural coherence during decoding and reduces the generation quality of semantic IDs. 3) Necessity of Prefix-Aware Loss: Eliminating Prefix-Aware Loss decreases the accuracy of coarse-grained token prediction and hinders generalization.

5.2.3 Online A/B Testing. To validate the real-world efficacy of LSIG, we deploy it on the Alibaba mobile app, serving over 12 million Daily Active Users (DAUs). In the domain targeted by our proposed strategy, LSIG achieves a 0.15% improvement in conversion and a 4.15% increase in GMV. These results indicate that LSIG not only enhances recommendation quality in offline evaluations but also delivers measurable business gains in real-world production environments.

6 Further Analysis

6.1 Prefix Sensitivity of TPA-Finetuned ID Embeddings

To assess the effectiveness of prefix-aware modeling in our Token Personalized Attention (TPA) framework, we analyze how the token embeddings evolve after finetuning. In particular, we examine the similarity of token embeddings at the same position across different items and investigate its relationship with prefix divergence. We define prefix divergence between two items as:

$$\text{Prefix Divergence} = 1 - \frac{\text{Length of common prefix}}{\text{Total prefix length}} \quad (15)$$

In conventional semantic ID models without TPA, token embeddings are position-dependent but context-independent. For instance, a token k at position 8 always yields the same embedding, regardless of the preceding tokens. In contrast, enabling TPA renders the token prefix-sensitive, as its embedding is dynamically modulated by the preceding sequence, even with the token and position held constant. To quantify this effect, we compute the cosine similarity of token embeddings at positions $k \in 4, 8, 16$ under varying levels of prefix divergence. As shown in Figure 6, token similarity consistently decreases as prefix divergence increases, indicating that TPA successfully modulates token representations according to prefix information.

These findings confirm our core design intuition: prefix information plays a pivotal role in modeling the semantics of hierarchical token sequences. By incorporating a prefix-aware transformation mechanism, TPA generates more fine-grained and distinguishable token embeddings—an essential capability for accurately modeling items in recommendation systems.

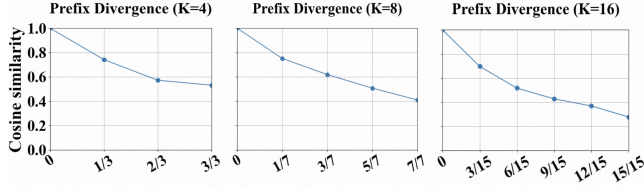


Figure 6: Cosine similarity of token embeddings across different levels of prefix divergence after TPA finetuning.

Table 3: Computational ablation of the TPA module. The best scores are highlighted in bold, and the second-best are underlined. LSIG is configured with two Decoder layers and one TPA module (Decoder $\times 2$ + TPA $\times 1$).

Variants	R@100	R@1000	N@100	N@1000
decoder*2	0.0574	0.1461	0.0164	0.0274
decoder*3	<u>0.0591</u>	<u>0.1509</u>	<u>0.0169</u>	<u>0.0282</u>
LSIG	0.0630	0.1599	0.0176	0.0295

6.2 Computation Efficiency Ablation of TPA

To assess the computational cost introduced by the TPA module, we analyze its theoretical complexity and compare its runtime efficiency against standard decoder configurations. Unlike standard self-attention layers that operate over the entire user behavior sequence, TPA restricts attention computation to tokens within a single item. This results in a significantly reduced computational complexity. Formally, standard self-attention scales as $O(N^2d)$, while the total complexity of all TPA amounts to $O(LNd)$, where N is the sequence length, L is the number of semantic ID tokens per item, and d is the hidden size. Since $L \ll N$, TPA imposes substantially lower computational overhead than a full-sequence decoder. To further quantify the efficiency-performance trade-off, we conduct an ablation study comparing three architectural variants: (1) COBRA (decoder $\times 2$), (2) COBRA (decoder $\times 3$), and (3) LSIG (decoder $\times 2$ + TPA $\times 1$).

As shown in Table 3, the results indicate that even when one decoder layer is replaced with a computationally efficient TPA block, the model consistently achieves superior performance across all metrics. This demonstrates that TPA not only maintains a low computational footprint but also enhances the model’s expressiveness, making it a practical and effective component for real-world deployment.

6.3 Scalability Analysis of HMTF Depth

To evaluate how the depth of the Hierarchical Multi-Token Prediction (HMTF) module influences performance, we increase the prediction depth $next^k$ from 1 to 15 and evaluate Recall@1000 and NDCG@1000. The results are shown in Figure 7 (Left). A clear scalability pattern is observed: as the HMTF depth increases, performance improves significantly up to depth 4, beyond which further gains diminish or slightly reverse. We attribute this to two main factors: 1. Semantic Relevance Decay: As the prediction horizon

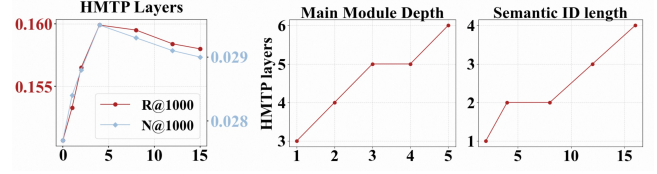


Figure 7: (Left): Performance trend of HMTF with increasing prediction depth (ID length = 16; Main module depth = 2). (Middle, Right): Optimal HMTF layer configurations under varying semantic ID lengths and main module depths.

extends (e.g., $next^{15}$), the semantic correlation between long-range tokens weakens, reducing modeling benefit. 2. Model Capacity Limitation: The backbone network used in our experiments has only two decoder blocks, which may be insufficient to capture and propagate long-range dependencies introduced by deeper HMTF layers.

As shown in Figure 7 (Middle and Right), to further validate the above analysis, we conduct extended experiments across different combinations of semantic ID lengths 2, 4, 8, 16 and backbone network depths 1, 2, 3, 4, 5, aiming to identify the optimal number of HMTF layers. Results indicate that longer semantic ID sequences benefit from deeper HMTF structures. Moreover, increasing the capacity of the backbone network effectively alleviates the bottleneck caused by limited HMTF depth. These findings validate the scalability and effectiveness of HMTF when appropriately aligned with model capacity and semantic ID length.

7 Conclusion

In this work, we propose LSIG, a generative recommendation framework leveraging long semantic IDs to represent items as sequences of hierarchical semantic tokens, thereby significantly enhancing the expressive capacity of recommendation models. We introduce the Token Personalized Attention (TPA) mechanism to effectively mitigate prediction confusion arising from semantic token ambiguity. During training, we design the Hierarchical Multi-Token Prediction (HMTF) module to explicitly model hierarchical semantic dependencies among tokens, incorporating a throttling gate mechanism to ensure consistency between training and inference phases. Additionally, we propose a Prefix-Aware Loss that dynamically emphasizes crucial prefix positions within semantic IDs, guiding the model to prioritize learning coarse-grained semantic structures, which leads to faster convergence and improved generalization. Extensive experiments demonstrate that LSIG achieves superior performance across various offline metrics and online scenarios. By effectively leveraging the hierarchical semantics of long semantic IDs, LSIG offers a robust and scalable solution for large-scale recommendation tasks.

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